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BUILDING A SIEM DASHBOARD USING ELK STACK

DIWAN SHAH FAHAD KHAN

DECLARATION

I hereby declare that the project entitled “**Building a SIEM Dashboard Using ELK Stack**” is my original work conducted as part of the **B.Tech – Computer Science and Engineering** program. The work presented in this report has been completed under the guidance of **ABHISHEK JYANT**.

I further declare that this project report has not been submitted previously, in whole or in part, for the award of any other degree, diploma, or qualification.

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ABSTRACT

This project focuses on the development of a Security Information and Event Management (SIEM) dashboard using the ELK Stack, consisting of Elasticsearch and Kibana, along with an Elastic log collection component. The main objective of the project is to collect, analyze, visualize, and monitor system authentication events in a centralized security monitoring environment.

The project was implemented using Kali Linux as a test environment. Authentication and SSH-related events were collected and analyzed in Kibana. A detection rule named “**SSH Brute Force Detection**” was configured to identify more than five matching authentication events within a five-minute period. During testing, multiple failed SSH login attempts were generated using a controlled local test environment. The rule successfully detected six matching events and generated an active security alert.

The triggered alert was investigated using authentication of event fields such as event action, username, source IP, event outcome, event category, and event dataset. The investigation confirmed the test activity as failed SSH authentication attempts from the local host. After the matching activity stopped, the alert successfully changed to a **Recovered** state.

A Kibana security dashboard containing **11 panels** was also developed to provide a visual overview of authentication activity, failures, users, source IPs, trends, and active security alerts. The project demonstrates how an ELK-based SIEM can be used for centralized log monitoring, security event detection, investigation, and visualization.

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1. INTRODUCTION

1.1 BACKGROUND

In modern computing environments, many security-related events are generated continuously by operating systems, applications, network devices, and authentication services. Monitoring these events manually can be difficult because security information is distributed across various sources. A Security Information and Event Management (SIEM) system helps organizations collect, centralize, analyze, and visualize security events to identify potentially suspicious activities.

The ELK Stack provides a flexible platform for centralized log management and security monitoring. Elasticsearch is used for storing and searching log data, while Kibana provides tools for log analysis, visualization, dashboards, and security monitoring. An Elastic log collection component can be used to collect events from monitored systems and forward them to Elasticsearch.

In this project, an ELK-based SIEM environment was implemented using Kali Linux as the test system. Authentication and SSH-related events were collected and analyzed through Kibana. A security monitoring dashboard was developed to visualize authentication activity, failures, users, source IP information, and security alerts. A detection rule was also configured to identify repeated SSH authentication failures and generate a security alert for investigation.

1.2 Problem Statement

Traditional log monitoring methods often require administrators or security teams to inspect individual log files manually. This approach becomes difficult when many authentication events are generated continuously. Important security events may be overlooked, and identifying repeated failed login attempts can take considerable time.

Another challenge is the lack of centralized visibility. Without a proper monitoring and visualization system, it can be difficult to understand authentication trends, identify frequently targeted users or source addresses, and determine whether repeated failures represent potentially suspicious activity.

Therefore, this project addresses the need for a centralized security monitoring solution that can collect authentication logs, analyze security events, detect repeated SSH authentication failures, generate alerts, and present the information through an easy-to-understand dashboard.

1.3 Objectives

1. To implement a centralized SIEM environment using the ELK Stack.
2. To collect and analyze system authentication and SSH-related logs.
3. To monitor successful and failed authentication events through Kibana.
4. To create visualizations and a centralized security monitoring dashboard.
5. To configure a detection rule for identifying repeated SSH authentication failures.
6. To generate and investigate security alerts based on detected suspicious activity.
7. To verify the alert lifecycle from **Active** to **Recovered**.
8. To provide a centralized and user-friendly view of security events for monitoring and investigation.

1.4 Scope of the project

The scope of this project is to develop a basic security monitoring and event detection environment using the ELK Stack. The project focuses primarily on system authentication and SSH-related events generated in a controlled Kali Linux environment.

The implemented system provides centralized log collection, searching, visualization, security alert generation, and basic incident investigation capabilities. The dashboard presents authentication activity through multiple visualizations and a recent event table, while the SSH Brute Force Detection rule identifies repeated authentication failures within a defined period.

The project is intended as an educational and practical demonstration of SIEM concepts. It can be extended in the future to monitor multiple systems, network traffic, application logs, malware-related events, and additional security threats.

2. TOOLS AND TECHNOLOGIES

2.1 Kali Linux

Kali Linux is a Debian-based Linux distribution designed for cybersecurity, penetration testing, digital forensics, and security analysis. In this project, Kali Linux was used as the test environment for generating and analyzing SSH authentication events. Failed SSH login attempts were generated in a controlled local environment to evaluate the SIEM detection mechanism.

2.2 Docker

Docker is a platform used to package and run applications in isolated containers. Containers provide a consistent environment in which applications and their dependencies can run without requiring a separate full operating system. In this project, Docker was used to deploy and manage the ELK components in a controlled environment.

Docker Compose was used to define and manage the Elasticsearch and Kibana services together. This simplified the deployment of the SIEM environment and made it easier to start, stop, and manage the required services.

2.3 Elasticsearch

Elasticsearch is a distributed search and analytics engine used to store, index, and search for large volumes of structured and unstructured data. It provides fast search capabilities and allows security events to be queried based on different fields and conditions.

In this project, Elasticsearch was used as the central storage and search engine for authentication and security event data collected from the Kali Linux environment. The collected events were indexed in Elasticsearch and accessed through Kibana for analysis, visualization, and security monitoring.

2.4 Kibana

Kibana is a web-based visualization and analysis platform that works with Elasticsearch. It provides an interface for searching, analyzing, visualizing, and monitoring data stored in Elasticsearch. Kibana also provides features for creating dashboards, detection rules, and security alerts.

In this project, Kibana was used to analyze authentication events through Discover, create security visualizations, build the SIEM dashboard, configure the SSH Brute Force Detection rule, and investigate generated security alerts. The final dashboard contains 11 panels that provide a centralized view of authentication activity and security events.

2.5 Elastic Agent

Elastic Agent is a unified agent provided by Elastic for collecting and monitoring data from systems and applications. It can collect several types of logs and metrics and forward them to Elasticsearch for centralized analysis.

In this project, Elastic Agent was used as the log collection component for collecting system authentication events from the Kali Linux environment. The collected events were sent to Elasticsearch and analyzed through Kibana. Authentication events were available under the **system.auth** datasets were used for visualization and SSH brute-force detection.

2.6 SSH and PAM Authentication

Secure Shell (SSH) is a network protocol used for securing remote login and communication with a system. Authentication events generated by SSH can provide useful security information, particularly when repeated login failures occur.

PAM (Pluggable Authentication Modules) is a framework used by Linux systems to manage authentication. In this project, SSH and PAM authentication events were monitored through the system authentication logs. These events provided information such as the login action, username, authentication outcome, and remote host information. The events were analyzed in Kibana and used as the basis for detecting repeated failed SSH authentication attempts.

2.7 KQL (Kibana Query Language)

Kibana Query Language (KQL) is a query language used in Kibana to search and filter data stored in Elasticsearch. It provides an effortless way to find specific events based on fields and values.

In this project, KQL was used to filter authentication events and investigate failed SSH login attempts. For example, authentication failures could be identified using fields such as

event.category and **event.outcome**. KQL was also useful while investigating the events associated with the SSH Brute Force Detection rule.

An example query used during analysis was:

event.category:"authentication" AND event.outcome:"failure"

This query filters the data to show authentication events whose outcome was a failure.

3. SYSTEM ARCHITECTURE

3.1 System Architecture Overview

The proposed SIEM environment is designed to collect system authentication events from a Kali Linux test system, process and store the collected data using the Elastic Stack and provide centralized monitoring through Kibana. The architecture consists of the log source, log collection component, Elasticsearch, Kibana, detection rules, security alerts, and the SIEM dashboard.

In the implemented environment, authentication and SSH-related events generated on the Kali Linux system are collected and forwarded to Elasticsearch. Kibana is then used to search and analyze these events, create visualizations, monitor security activity, and configure detection rules. When the configured SSH brute-force condition is satisfied, Kibana generates a security alert that can be investigated and monitored until the alert is recovered.

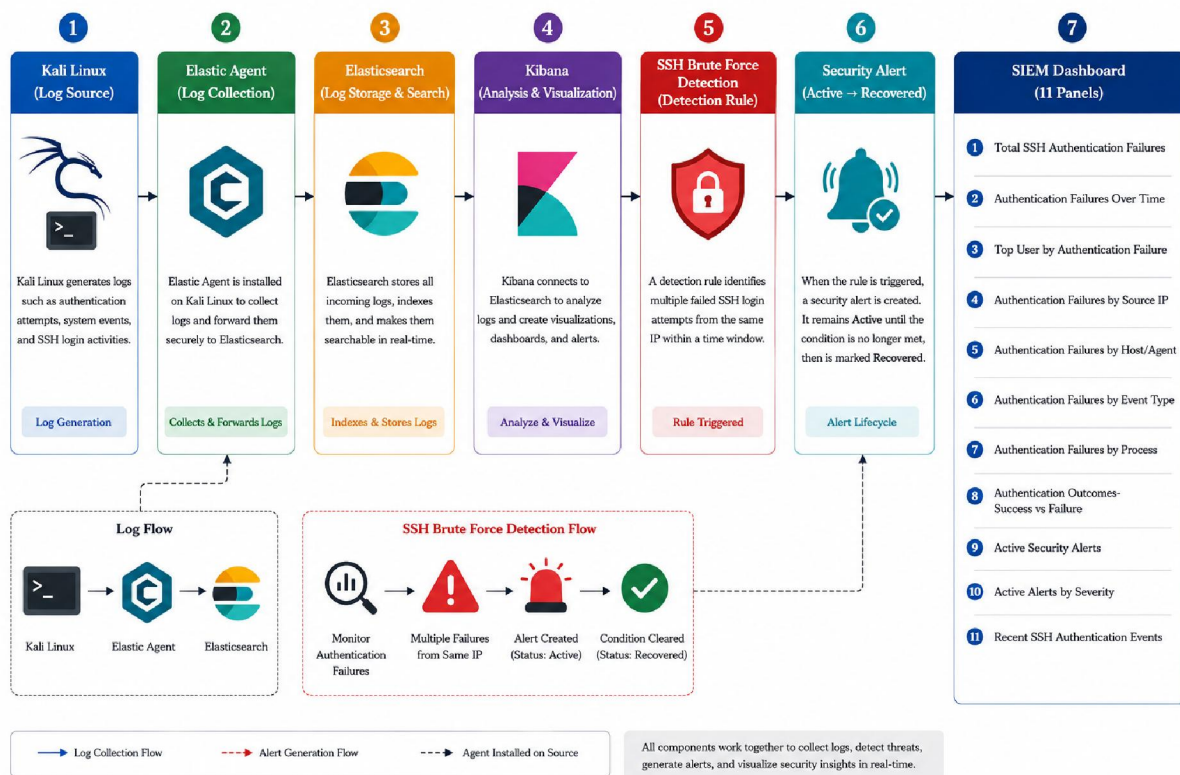


Figure 3.1: Architecture and Data Flow of the ELK Based SIEM.

3.2 Data Flow

The data flow in the implemented SIEM environment begins with authentication and SSH events generated on the Kali Linux system. These events are collected by the Elastic Agent and forwarded to Elasticsearch for centralized storage and analysis.

Elasticsearch indexes the collected events and makes them available in Kibana. In Kibana Discover, the authentication events can be searched and investigated using fields such as **event.action**, **user.name**, **source.ip**, **event.outcome**, **event.category**, and **event.dataset**.

The collected events are also evaluated by the **SSH Brute Force Detection** rule. The configured rule generates an alert when more than five matching authentication events are detected within five minutes. During the controlled test, six matching events were detected, and the alert became active.

The alert was then investigated using the related authentication events. The test events showed **ssh_login** as the event action, **wronguser** as the username, **::1** (**The source IP ::1 represents the IPv6 loopback address, indicating that the test connection originated from the local system**) as the source IP, and **failure** as the authentication outcome. After the matching activity stopped, the alert changed to the **Recovered** state.

Finally, the analyzed security information is presented through the Kibana SIEM dashboard, which contains 11 panels for monitoring authentication failures, trends, users, source information, authentication outcomes, and security alerts.

3.3 System Components

The implemented SIEM environment consists of several interconnected components that work together to collect, process, analyze, and monitor security events. Each component performs a specific function in the overall security monitoring process.

3.3.1 Kali Linux

Kali Linux was used as the test environment for generating SSH authentication activity. Controlled failed login attempts were created to verify whether the configured detection rule could identify repeated authentication failures.

3.3.2 Elastic Agent

Elastic Agent was used as the log collection component. It collected authentication-related events from the Kali Linux system and forwarded them to Elasticsearch for centralized storage and analysis.

3.3.3 Elasticsearch

Elasticsearch served as the central storage and search engine for the collected security events. The authentication logs were indexed in Elasticsearch, allowing them to be searched and analyzed through Kibana.

3.3.4 Kibana

Kibana provided the interface for analyzing the collected events and monitoring security activity. It was used for Discover-based investigation, visualization creation, dashboard development, detection rule configuration, and security alert investigation.

3.3.5 SSH Brute Force Detection Rule

The SSH Brute Force Detection rule was configured to identify more than five matching authentication events within a five-minute period. During testing, six failed matching SSH authentication events caused the rule to generate an active security alert.

3.3.6 Security Alert

Security alerts were used to indicate when the configured detection condition was satisfied. The generated SSH brute-force alert was investigated using the associated authentication events and later changed to a recovered state after the matching activity stopped.

3.3.7 SIEM Dashboard

The Kibana SIEM dashboard provides a centralized visual representation of authentication and security activity. The final dashboard contains 11 panels, including authentication failure metrics, trends, source information, user information, authentication outcomes, active security alerts, alert severity, and recent SSH authentication events.

3.4 Architecture Summary

The implemented architecture provides a complete flow from security event generation to centralized monitoring and investigation. Authentication events generated on the Kali Linux system are collected by Elastic Agent and stored in Elasticsearch. Kibana provides the interface

for searching and analyzing these events, while the configured detection rule identifies repeated SSH authentication failures.

When the detection condition is satisfied, a security alert is generated and can be investigated using the related event information. The resulting security information is presented through the Kibana SIEM dashboard, providing a centralized view of authentication activity and detected security events. This architecture demonstrates the basic working process of a SIEM system in a controlled test environment.

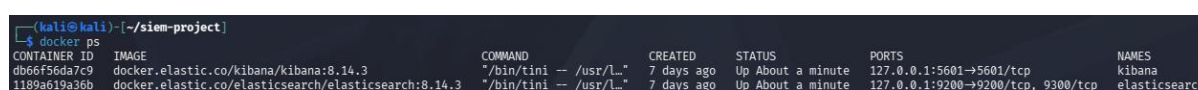
4. IMPLEMENTATION

4.1 ELK Stack Environment Setup

The ELK-based SIEM environment was implemented using Docker to provide an isolated and manageable environment for the required services. The main components used in the setup were Elasticsearch and Kibana. Elasticsearch was configured as the central storage and search engine, while Kibana was used for data analysis, visualization, dashboard creation, and security monitoring.

Docker Compose was used to define the required services and their configurations. The Elasticsearch and Kibana services were connected through a common Docker network, allowing them to communicate with each other. Persistent storage was also configured for Elasticsearch so that indexed data could be retained when the container was restarted.

The environment was evaluated to ensure that the Elasticsearch and Kibana services were running correctly before proceeding with log collection and security monitoring.



```
(kali@kali) ~/-/siem-project
$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
db66f56da7c9	docker.elastic.co/kibana/kibana:8.14.3	"/bin/tini -- /usr/l..."	7 days ago	Up About a minute	127.0.0.1:5601->5601/tcp	kibana
1189a619a36b	docker.elastic.co/elasticsearch/elasticsearch:8.14.3	"/bin/tini -- /usr/l..."	7 days ago	Up About a minute	127.0.0.1:9200->9200/tcp, 9300/tcp	elasticsearch

Figure 4.1: ELK Stack Services Running in Docker.

4.2 Docker Configuration

Docker Compose was used to configure the ELK services required for the SIEM environment. The configuration defined Elasticsearch and Kibana as separate services and allowed them to communicate through a common Docker network.

Elasticsearch was configured as a single-node instance for the project environment. Security features were enabled to protect access to Elasticsearch. Persistent volume storage was configured to retain Elasticsearch data, while resource limits were applied to manage memory and CPU usage within the test environment.

The Docker Compose configuration simplified the management of the ELK services. The services could be started, stopped, and checked using Docker Compose commands, making the SIEM environment easier to maintain during development and testing.

```

GNU nano 9.1                                     docker-compose.yml
services:
  elasticsearch:
    image: docker.elastic.co/elasticsearch/elasticsearch:8.14.3
    container_name: elasticsearch

    environment:
      - discovery.type=single-node
      - xpack.security.enabled=true
      - xpack.security.enrollment.enabled=false
      - ELASTIC_PASSWORD=${ELASTIC_PASSWORD}
      - "ES_JAVA_OPTS=-Xms1g -Xmx1g"

    volumes:
      - es_data:/usr/share/elasticsearch/data

  kibana:
    image: docker.elastic.co/kibana/kibana:8.14.3
    container_name: kibana

    environment:
      - ELASTICSEARCH_HOSTS=http://elasticsearch:9200
      - ELASTICSEARCH_USERNAME=kibana_system
      - ELASTICSEARCH_PASSWORD=${KIBANA_PASSWORD}
      - XPACK_ENCRYPTEDSAVEDOBJECTS_ENCRYPTIONKEY=MySIEMProjectKibanaKey786

    ports:
      - "127.0.0.1:5601:5601"

    networks:
      - siem_network

    depends_on:
      - elasticsearch

networks:
  siem_network:
    driver: bridge

volumes:
  es_data:

```

Figure 4.2: Docker Compose Configuration for the ELK Environment

4.3 Elasticsearch Configuration

Elasticsearch was configured as the central data storage and search component of the SIEM environment. It received the authentication events collected from the Kali Linux system through Elastic Agent and indexed them for searching and analysis.

A single-node configuration was used because the project was implemented in a controlled local environment. Elasticsearch security features were enabled to provide authenticated access to the service. Persistent storage was configured so that the indexed data could be retained when the Elasticsearch container was restarted.

After configuration, the Elasticsearch service was started, and its status was checked through Docker. The service was required to be run correctly before Kibana and Elastic Agent could be used for further analysis.



CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
db66f56da7c9	docker.elastic.co/kibana/kibana:8.14.3	"/bin/tini -- /usr/L..."	7 days ago	Up About a minute	127.0.0.1:5601→5601/tcp	kibana
1189a619a36b	docker.elastic.co/elasticsearch/elasticsearch:8.14.3	"/bin/tini -- /usr/L..."	7 days ago	Up About a minute	127.0.0.1:9200→9200/tcp, 9300/tcp	elasticsearch

Figure 4.3: Elasticsearch Service Running in Docker Environment

4.4 Kibana Configuration

Kibana was configured as a web-based interface for analyzing and visualizing the data stored in Elasticsearch. After the Elasticsearch service was running, Kibana was started as part of the ELK environment and connected to Elasticsearch.

Kibana was used throughout the project to access the collected authentication events through Discover, create visualizations, configure security detection rules, investigate alerts, and develop the final SIEM dashboard.

The Kibana interface provided centralized access to the security monitoring functions required for the project. Authentication events from the **system.auth** dataset were analyzed in Kibana and used to create the dashboard and SSH brute-force detection workflow.

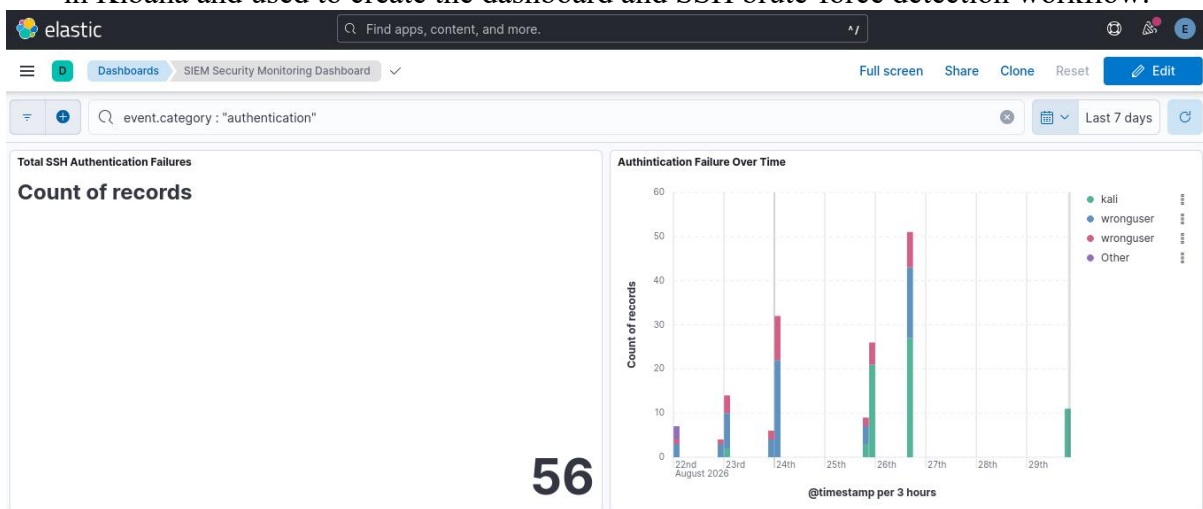


Figure 4.4: Kibana Interface Used for Security Monitoring

4.5 Elastic Agent and Log Collection

Elastic Agent was used to collect authentication-related logs from the Kali Linux system and forward them to Elasticsearch. The agent provided the connection between the log source and the centralized Elasticsearch storage used by the SIEM environment.

The collected events included SSH authentication activity generated on the Kali Linux system. These events contained useful fields such as event action, username, source IP, event outcome, event category, event dataset, and host information.

After the agent was configured and running, the collected events were verified in Kibana Discover. The **system.auth** dataset was used to identify and analyze authentication events, including failed SSH login attempts.

 event.action	ssh_login
 event.agent_id_status	verified
 event.category	authentication
 event.dataset	system.auth
 event.ingested	Aug 29, 2026 @ 20:02:33.000
 event.kind	event
 event.module	system
 event.outcome	failure
 event.timezone	+05:30
 source.ip	:::1
 source.port	44,254
 system.auth.ssh.event	Failed
 system.auth.ssh.method	password
 tags	system-auth
 user.name	wronguser

Figure 4.5: Authentication Events Collected by Elastic Agent

4.6 Authentication Log Analysis in Kibana

After the authentication logs were collected, Kibana Discover was used to analyze the events in detail. A KQL filter was applied to focus on authentication failures and reduce unrelated events from the analysis.

The authentication events were examined using important fields such as **event.action**, **user.name**, **source.ip**, **event.outcome**, **event.category**, **event.dataset**, and **host.hostname**. These fields helped identify the type of authentication activity, the affected user, the source of the connection, the authentication result, and the system that generated the event.

During the investigation, failed SSH login events were identified with **event.action** set to **ssh_login** and **event.outcome** set to **failure**. The test events also showed **wronguser** as the username and **::1** as the source IP. This analysis provided the event information required to validate the SSH brute-force detection rule.

4.7 SSH Brute Force Detection Rule

A detection rule was configured in Kibana to identify repeated failed SSH authentication attempts. The rule was named “**SSH Brute Force Detection**” and was designed to identify potentially suspicious authentication activity within a defined period.

The rule was configured to trigger when the number of matching events was greater than five within a five-minute period. This threshold was selected to demonstrate the detection of repeated SSH authentication failures in the controlled test environment.

During testing, six matching authentication events were detected within five minutes. As a result, the detection rule generated a security alert, and the alert status changed to **Active**. The generated alert was then investigated using the associated authentication events.

Feature	Stack management
Last updated	Aug 29, 2026 @ 23:33:16.787
Started	Aug 29, 2026 @ 20:03:47.876
Rule category	Elasticsearch query
Rule	SSH Brute Force Detection
Rule tags	—
Evaluation values	—
Evaluation threshold	5
Reason	Document count is 12 in the last 5m in .alerts-security.alerts-default,apm-*-transaction*,auditbeat-*,endgame-*,filebeat-*,logs-*,packetbeat-*,traces-apm*,winlogbeat-*,-*elastic-cloud-logs-* data view. Alert when greater than 5.

Figure 4.7: SSH Brute Force Detection Rule Configuration

4.8 Security Alert Generation and Investigation

To validate the SSH Brute Force Detection rule, a controlled authentication test was performed in the Kali Linux environment. Multiple failed SSH login attempts were generated using an invalid username. The resulting authentication events were collected and analyzed in Kibana.

During the test, six matching authentication failure events were detected within a five-minute period. Since the configured threshold was greater than five events, the detection rule generated a security alert and its status changed to **Active**.

The generated events were then investigated in Kibana to verify the details of the detected activity. The investigation showed **ssh_login** as the event action, **wronguser** as the **username**, **:::1** as the source IP, and **failure** as the authentication outcome. The event category was **authentication**, and the event dataset was **system.auth**, with the event generated by the Kali Linux host.

After the test activity stopped and no further matching events were detected, the alert status changed from **Active** to **Recovered**. This confirmed that the configured detection rule and alert lifecycle were functioning as expected in the controlled test environment.

Alerts

History

Columns

10

Sort fields

1

10 alerts

Fields

Updated 30 seconds ago

☐	Actions	Alert Status	Last updated ↓	Started	Rule category	Rule
☐	...	Active	Aug 30, 2026 @ 01:12:59.493	Aug 30, 2026 @ 01:12:59.493	Elasticsearch query	SSH Brute Force Detection
☐	...	Recovered	Aug 30, 2026 @ 01:11:56.413	Aug 30, 2026 @ 00:37:45.211	Elasticsearch query	SSH Brute Force Detection
☐	...	Recovered	Aug 30, 2026 @ 00:36:39.380	Aug 30, 2026 @ 00:26:14.224	Elasticsearch query	SSH Brute Force Detection

Figure 4.8.1: Active SSH Brute Force Alert

host.hostname	kali
kibana.alert.evaluation.conditions	Number of matching documents is greater than 5
kibana.alert.rule.producer	stackAlerts
kibana.alert.reason.text	Document count is 9 in the last 5m in .alerts-security.alerts-default,apm*-transaction*,auditbeat-*,endgame-*,filebeat-*,logs-*,packetbeat-*,traces-apm*,winlogbeat-*,-elastic-cloud-logs-* data view. Alert when greater than 5.
kibana.alert.rule.name	SSH Brute Force Detection
host.name	kali
event.kind	signal
host.id	246bd9421a9a4c0d9da68d5f5abcfabc
kibana.alert.title	rule 'SSH Brute Force Detection' matched query
kibana.alert.workflow_status	open

Figure 4.8.2: Investigation of the Detected SSH Authentication Failure

Alerts

History

Columns

10

Sort fields

1

10 alerts

Fields

Updated 30 seconds ago

☐	Actions	Alert Status	Last updated ↓	Started	Rule category	Rule
☐	...	Recovered	Aug 30, 2026 @ 01:21:19.047	Aug 30, 2026 @ 01:12:59.493	Elasticsearch query	SSH Brute Force Detection
☐	...	Recovered	Aug 30, 2026 @ 01:11:56.413	Aug 30, 2026 @ 00:37:45.211	Elasticsearch query	SSH Brute Force Detection
☐	...	Recovered	Aug 30, 2026 @ 00:36:39.380	Aug 30, 2026 @ 00:26:14.224	Elasticsearch query	SSH Brute Force Detection

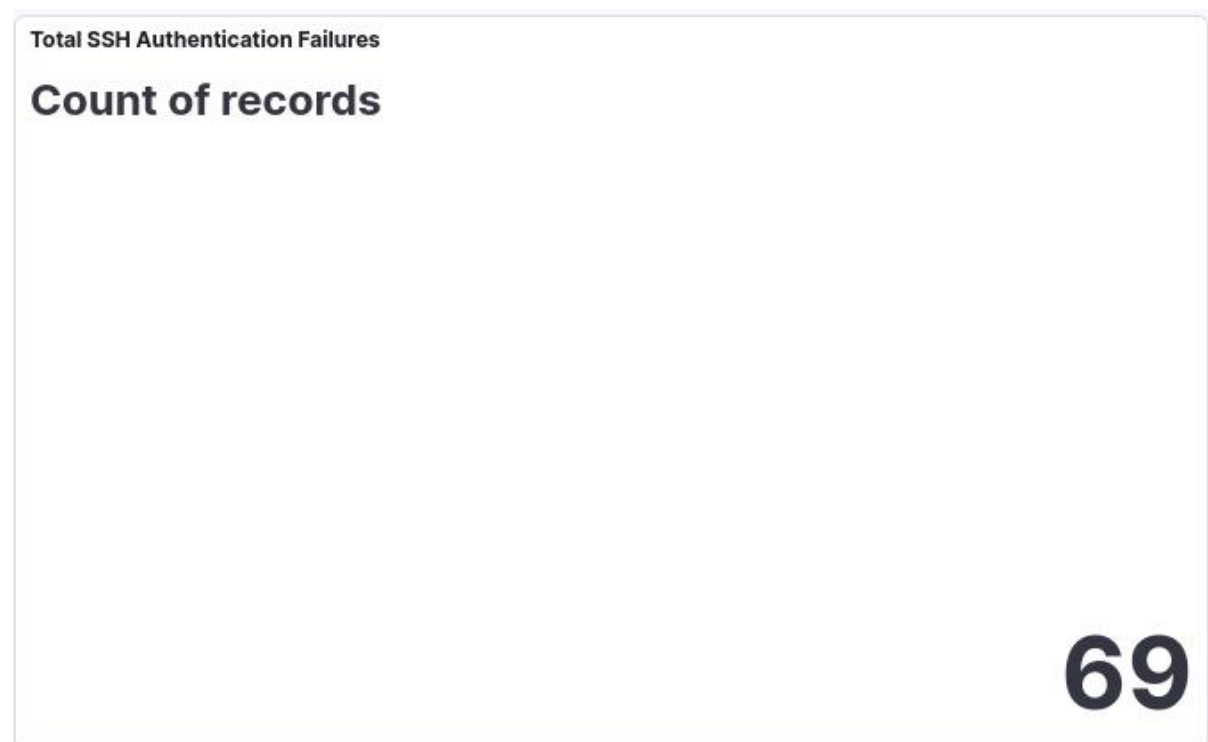
Figure 4.8.3: SSH Brute Force Detection Alert in Recovered State

4.9 SIEM Dashboard Implementation

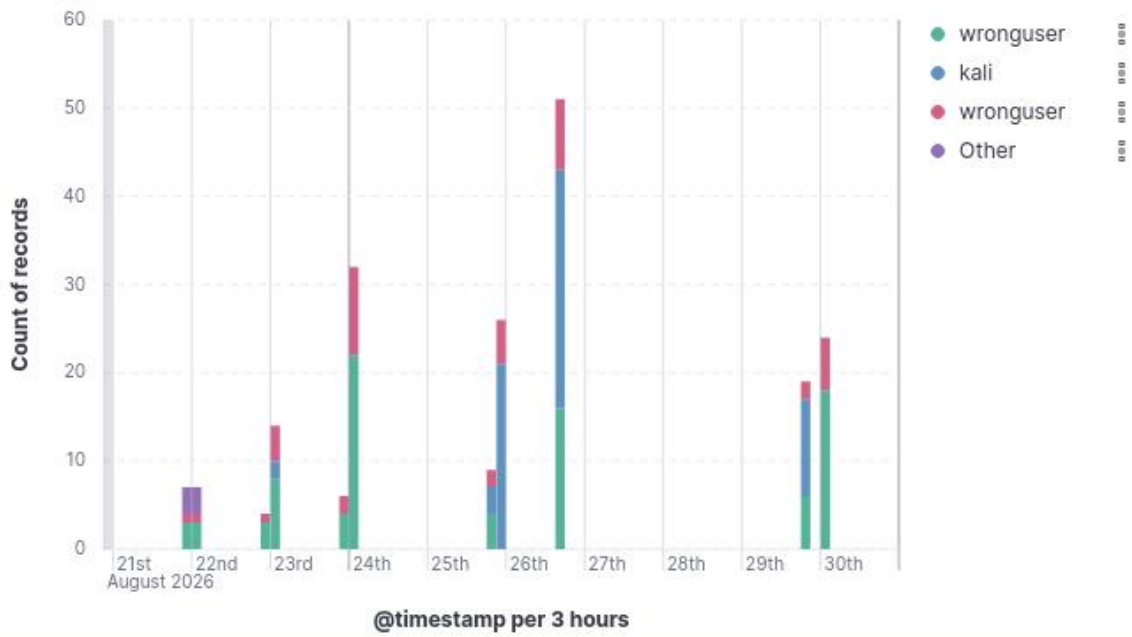
A centralized SIEM dashboard was developed in Kibana to provide a visual overview of authentication and security-related activity. The dashboard combines multiple visualizations and a recent authentication-events table to make important security information easier to monitor and analyze.

The final dashboard contains 11 panels covering authentication failures, authentication trends, users, source IP information, host and agent information, event types, processes, authentication outcomes, active security alert severity, and recent SSH authentication events.

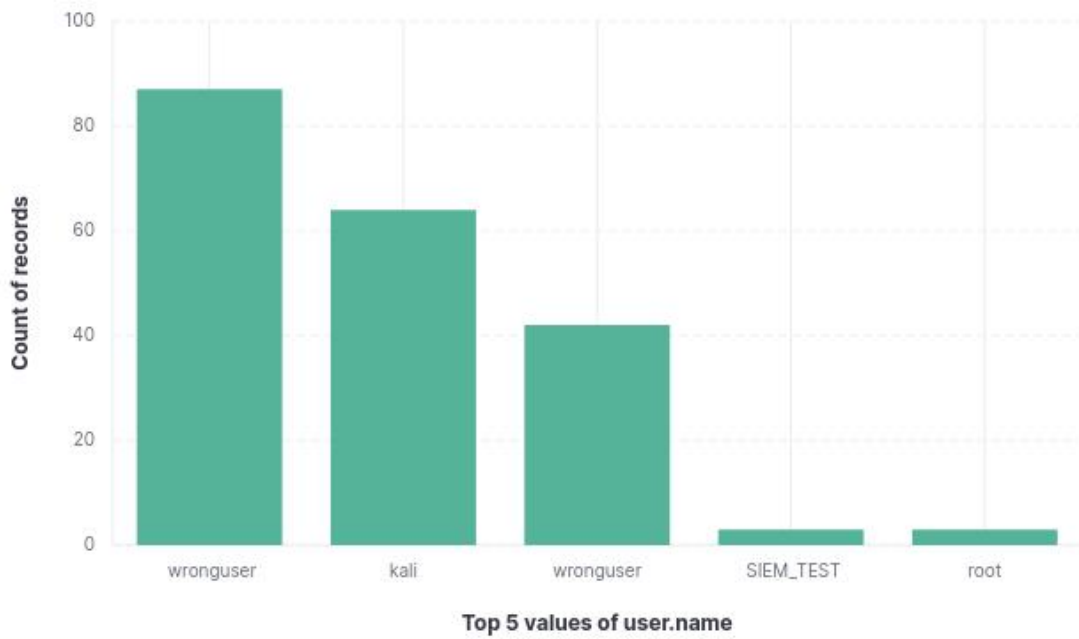
The dashboard provides a centralized monitoring view that helps identify authentication-related activity and supports further investigation of suspicious events. The visualizations can also be used to understand patterns and changes in authentication activity over the selected period.



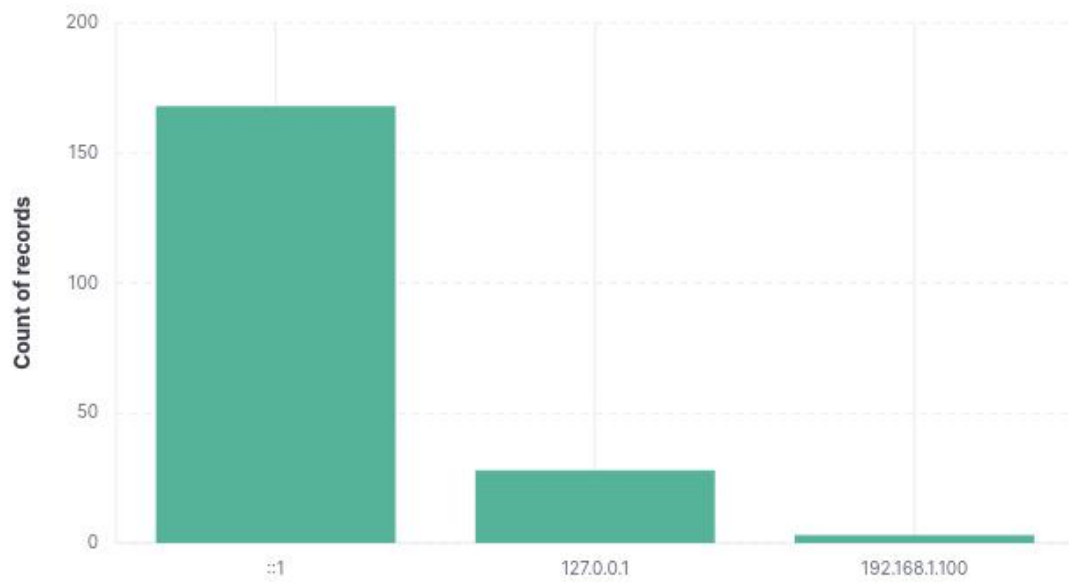
Authintication Failure Over Time



Top Users by Authentication Failure

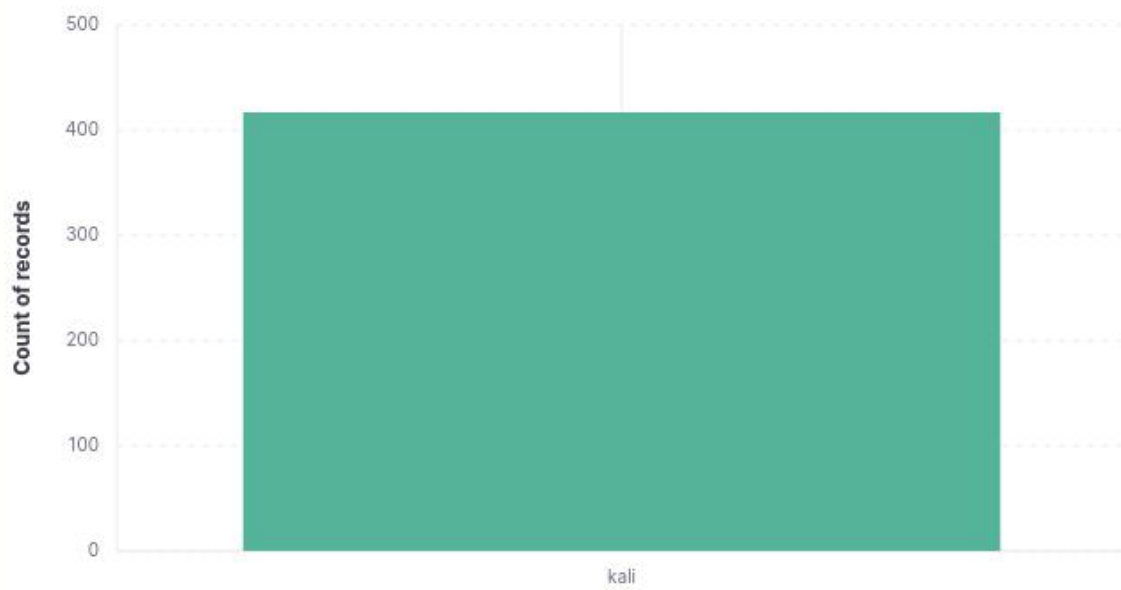


Authentication Failures by Source IP



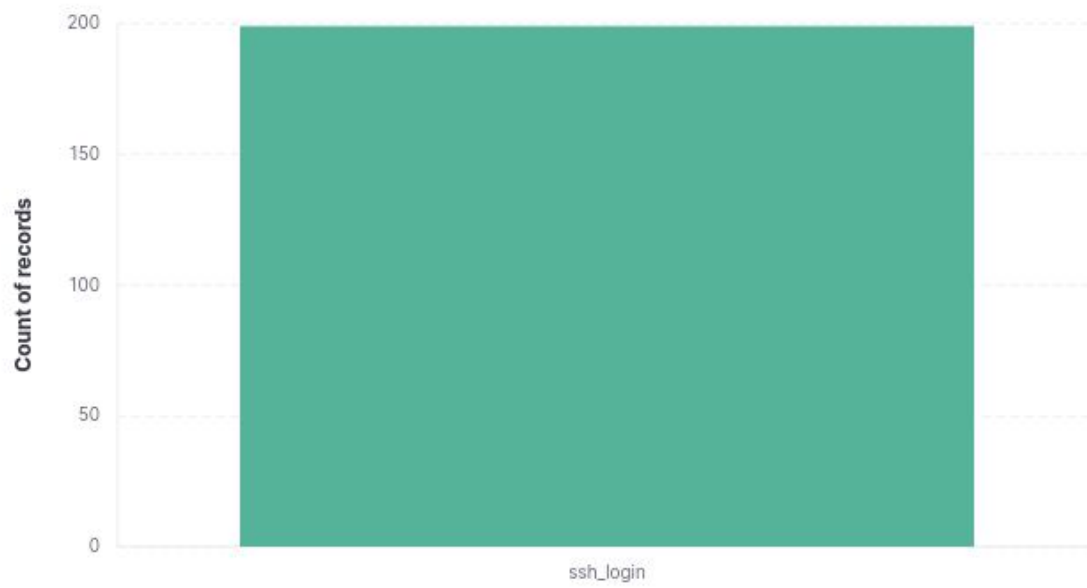
Top 5 values of source.ip

Authentication Failures by Host/Agent



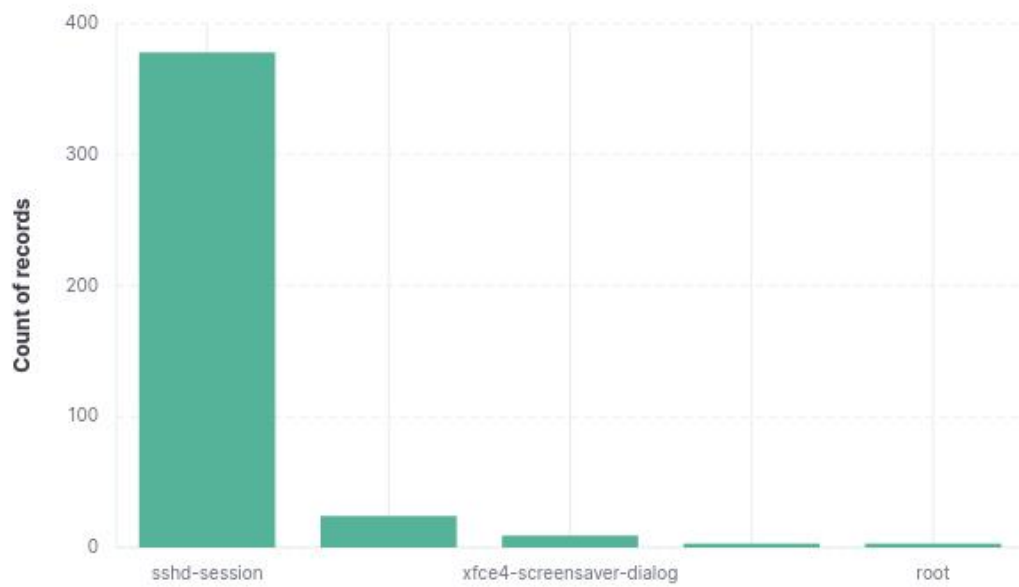
Top 5 values of agent.name

Authentication Failures by Event Type



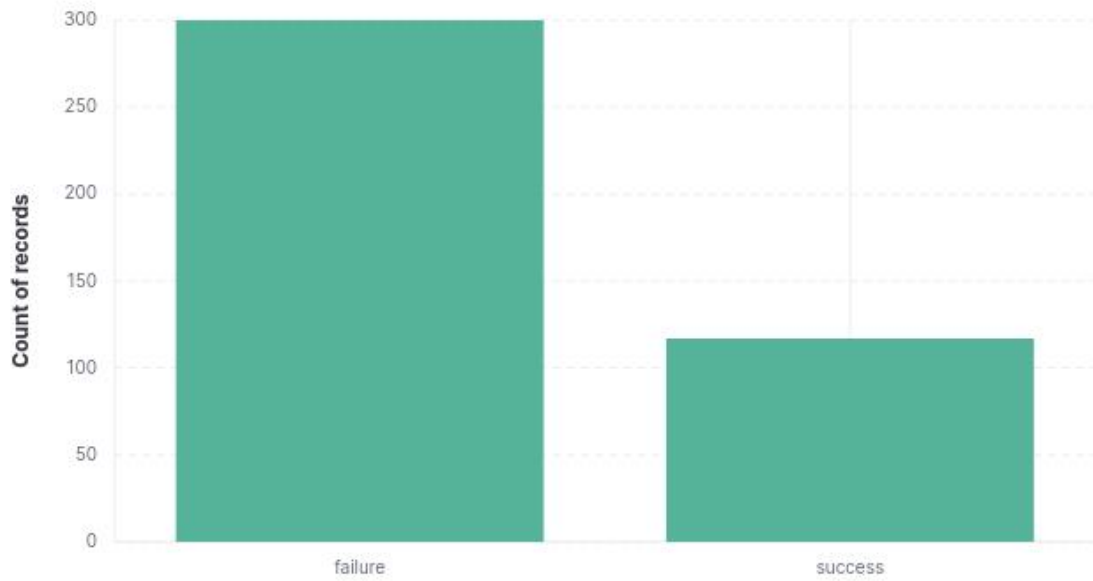
Top 5 values of event.action

Authentication Failures by Process



Top 5 values of process.name

Authentication Outcomes - Success vs Failure



Top 5 values of event.outcome

Active Security Alerts

Count of records

465



Figure 4.9: Final SIEM Dashboard with 11 Monitoring Panels

5. RESULT AND ANALYSIS

5.1 Authentication Log Collection Results

The implemented SIEM environment successfully collected authentication-related events from the Kali Linux test system. The events were received through the configured log collection setup and were made available in Kibana for searching and analysis.

The collected events included SSH authentication activity and provided useful fields such as event action, username, source IP, authentication outcome, event category, and event dataset. The presence of the **system.auth** dataset confirmed that authentication-related system events were being processed and made available for security monitoring.

During testing, failed SSH authentication events were successfully identified in Kibana. These events provided the required data for creating visualizations and evaluating the SSH Brute Force Detection rule.

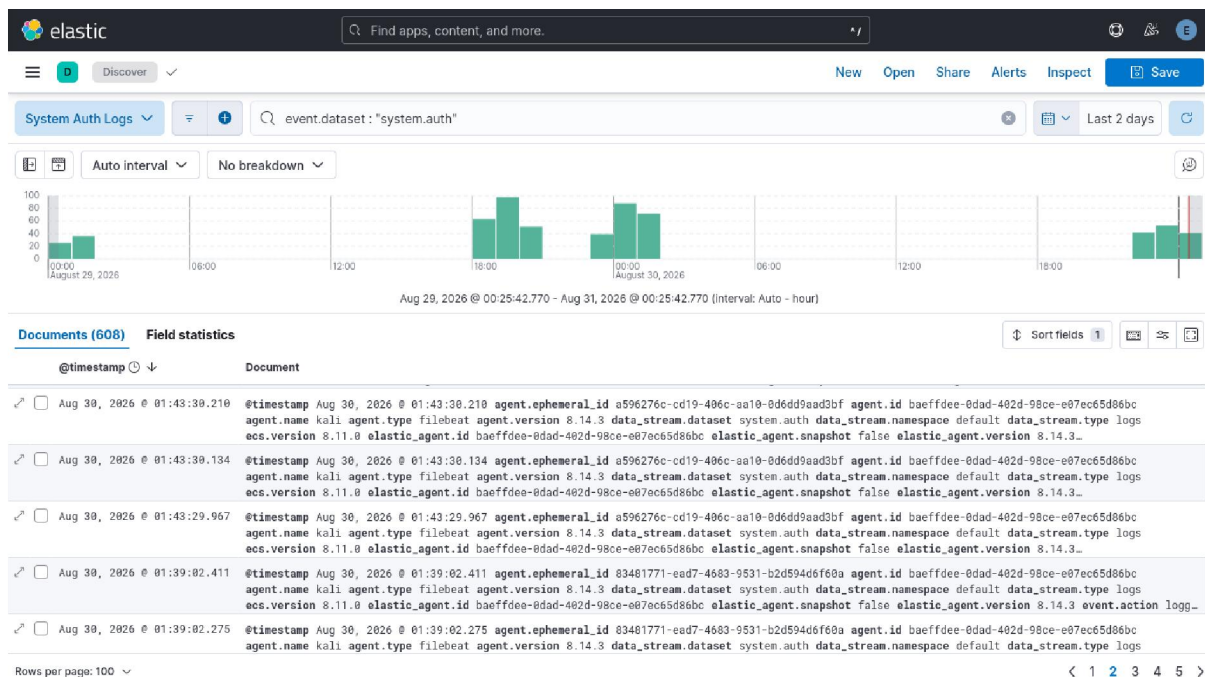


Figure 5.1: Authentication Events Collected and Displayed in Kibana Discover

5.2 SSH Authentication Failure Detection Results

The SIEM environment was evaluated by generating controlled failed SSH authentication attempts on the Kali Linux system. The purpose of this test was to verify whether.

The collected authentication events could be identified and used for security monitoring.

The failed authentication events were successfully recorded in Kibana with relevant information such as `event.action`, `user.name`, `source.ip`, and `event.outcome`. The test events showed `ssh_login` as the event action, `wronguser` as the username, `::1` as the source IP, and `failure` as the authentication outcome.

These events demonstrated that the SIEM environment was successfully collecting and identifying failed SSH authentication activity. The collected events were subsequently used to evaluate the configured SSH Brute Force Detection rule.

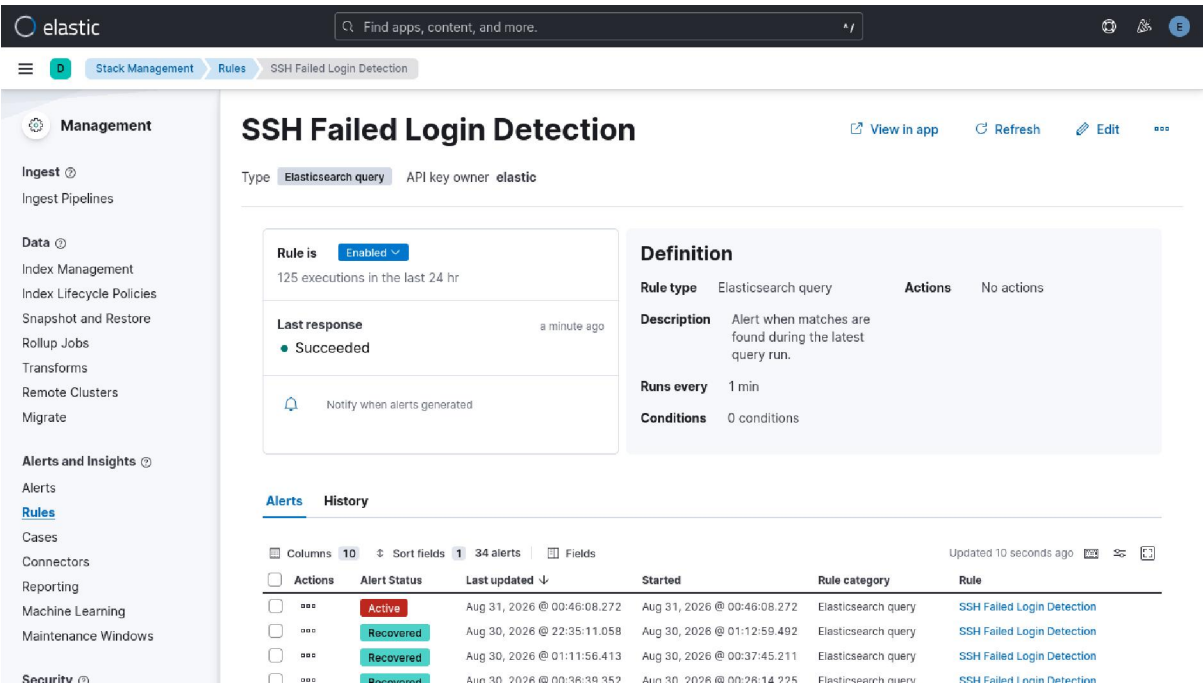


Figure 5.2: Failed SSH Authentication Event Identified in Kibana

5.3 Alert Investigation and Recovery Results

After the SSH brute-force detection rule generated an active alert, the alert was investigated using the associated authentication activity. The alert provided information about the detection rule, alert status, timestamp, and events that satisfied the configured detection condition.

The generated alert was observed in the active state while the failed authentication activity was occurring. After the matching authentication activity stopped and the detection condition was no longer satisfied, the alert changed from **Active** to **Recovered**.

This test confirmed that the configured detection workflow was able to identify repeated SSH authentication failures, generate a security alert, and subsequently update the alert status when the suspicious activity stopped.

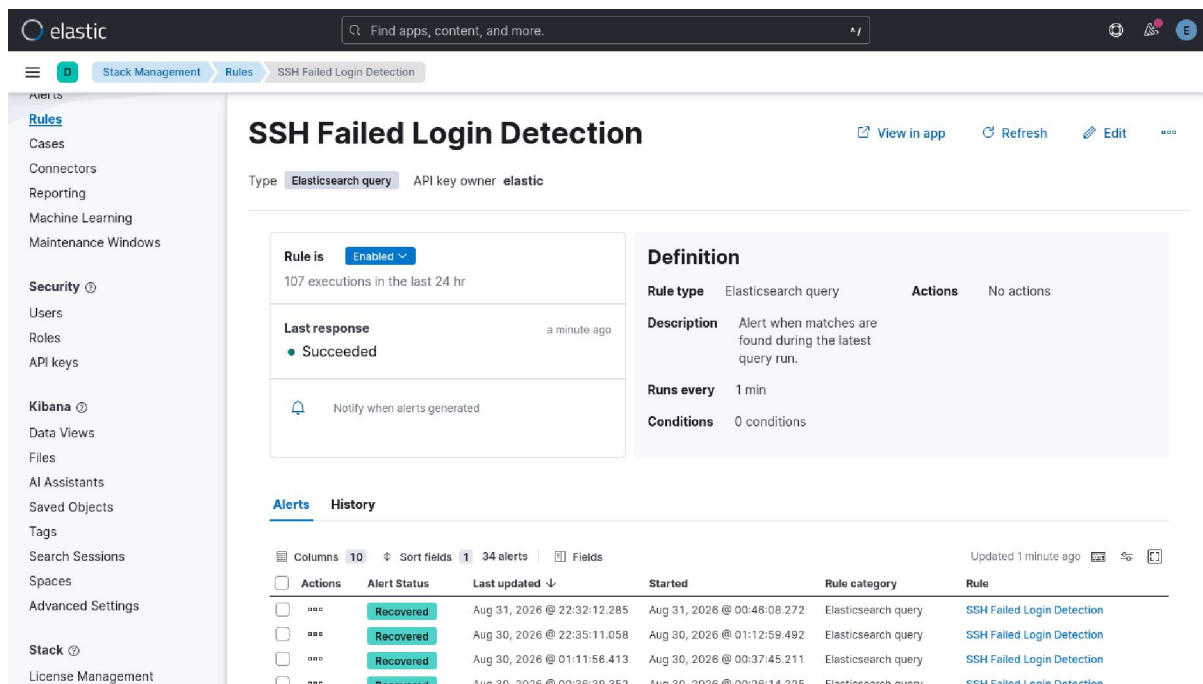


Figure 5.3: Recovered SSH Brute Force Detection Alert

5.4 Dashboard Analysis

The final SIEM dashboard was developed in Kibana to provide a centralized view of authentication activity and security events. The dashboard contains 11 panels that present various aspects of the collected authentication data and detected security activity.

The dashboard includes visualizations for total SSH authentication failures, authentication failures over time, top users by authentication failure, authentication failures by source IP, host or agent, event type, and process. It also provides a comparison of successful and failed authentication outcomes.

In addition to the authentication of visualizations, the dashboard contains panels for active security alerts and active alerts by severity. A recent SSH authentication events table provides detailed information about recent events and supports further investigation.

The combination of these panels provides a centralized monitoring view and makes it easier to identify authentication trends, repeated failures, suspicious activity, and active security alerts within the test environment.

5.4.1 Analysis of Dashboard Panels

1. Total SSH Authentication Failures

This panel displays the total number of failed SSH authentication attempts detected in the selected time range. It provides a quick overview of the authentication of failure activity and helps identify periods with an unusually high number of failed login attempts.

2. Authentication Failures Over Time

This panel shows the variation in SSH authentication failures over time. The time-based view helps identify sudden increases or repeated patterns of failed authentication attempts, which can be useful for detecting suspicious login activity.

3. Top User by Authentication Failure

This panel identifies the users associated with the highest number of failed authentication attempts. It helps in identifying accounts that may be targeted by repeated login attempts or may require further investigation.

4. Authentication Failures by Source IP

This panel shows the distribution of authentication failures based on the source IP addresses recorded in the authentication events. It helps identify the sources associated with repeated failed login attempts and supports investigation of suspicious authentication activity.

5. Authentication Failures by Host/Agent

This panel shows the authentication of failures associated with different hosts or Elastic Agents. It helps verify which monitored system is generating authentication-related activity and provides a useful overview of the distribution of events across monitored hosts.

6. Authentication Failures by Event Type

This panel categorizes authentication failures according to their event types. It helps in understanding the nature of the recorded authentication activity and provides an additional way to analyze the events collected by the SIEM environment.

7. Authentication Failures by Process

This panel displays authentication failures grouped by the process responsible for generating recorded events. It helps identify which system process is associated with the authentication activity and provides additional context during security event investigation.

8. Authentication Outcomes -- Success vs Failure

This panel compares successful and failed authentication outcomes recorded by the SIEM. It provides a quick view of the overall authentication status and helps identify whether the monitored system is experiencing an increased number of failed authentication attempts.

9. Active Security Alerts

This panel displays security alerts that are currently active in the SIEM environment. It provides a quick indication of ongoing detected security activity and helps the analyst identify alerts that require investigation.

10. Active Alerts by Severity

This panel categorizes active security alerts according to their severity level. It helps prioritize the investigation of alerts by providing an overview of the severity associated with detected security events.

11. Recent SSH Authentication Events

This table displays recent SSH authentication events along with relevant event information. It provides detailed evidence that can be used to investigate individual authentication attempts and understand the activity behind detected security alerts.

5.5 Overall Results

The implementation and testing of the ELK-based SIEM environment produced the expected results. Authentication events generated on the Kali Linux system were successfully collected and made available in Kibana for analysis. The collected data provided the required information for monitoring SSH authentication activity and identifying failed login attempts.

The SSH Brute Force Detection rule successfully detected repeated failed authentication attempts. During the controlled test, six matching events were generated within the configured five-minute period, causing the security alert to enter the Active state. The alert was investigated using the associated event information and later changed to the Recovered state after the matching activity stopped.

The final Kibana dashboard provided a centralized view of authentication activity through 11 monitoring panels. These panels supported analysis of authentication failures, trends, users, source information, event types, processes, authentication outcomes, and security alerts.

Overall, the results demonstrate that the implemented SIEM environment can collect authentication logs, detect repeated SSH authentication failures, generate security alerts, support investigation, and present security information through a centralized monitoring dashboard.

6. CONCLUSION

6.1 Conclusion

The project successfully demonstrated the implementation of a Security Information and Event Management (SIEM) environment using the ELK Stack. Elasticsearch was used for centralized storage and searching of security events, while Kibana provided the interface for event analysis, visualization, alert monitoring, and dashboard development. Elastic Agent was used to collect authentication-related events from the Kali Linux system.

The implemented environment successfully detected repeated SSH authentication failures using the configured SSH Brute Force Detection rule. During controlled testing, six matching authentication events within the configured five-minute period triggered an active security alert. The alert was investigated using the associated event information and subsequently changed to the Recovered state after the matching activity stopped.

The final SIEM dashboard provided a centralized view of authentication and security activity through 11 monitoring panels. The project therefore demonstrated the complete basic SIEM workflow of **log collection, centralized storage, event analysis, threat detection, alert generation, investigation, and security monitoring**.

Overall, the project provided practical experience in implementing and operating an ELK-based SIEM environment and demonstrated how centralized security monitoring can help identify and investigate suspicious authentication activity.

6.2 Project Outcomes

The major outcomes achieved through the project are as follows:

1. **ELK Environment Implementation:** An ELK-based SIEM environment was successfully deployed using Docker.
2. **Centralized Log Collection:** Authentication-related events from the Kali Linux system were collected using Elastic Agent and made available in Elasticsearch.
3. **Authentication Event Analysis:** SSH authentication events were analyzed in Kibana using relevant event fields and KQL queries.
4. **SSH Brute Force Detection:** A detection rule was configured to identify repeated SSH authentication failures within a defined period.

5. **Security Alert Generation:** Controlled testing successfully triggered an active security alert after six matching authentication events were detected within five minutes.
6. **Alert Investigation:** The alert generated was investigated using the associated authentication of event information, including the username, source IP, event action, and authentication outcome.
7. **Alert Recovery:** The alert successfully changed to the Recovered state after the matching suspicious activity stopped.
8. **SIEM Dashboard Development:** A centralized Kibana dashboard containing 11 monitoring panels was developed to provide an overview of authentication activity and security alerts.
9. **Practical Security Monitoring:** The project provides practical experience with log collection, event analysis, detection rules, security alerts, investigation, and SIEM dashboard development.

6.3 Future Scope

The current project was implemented in a controlled local environment and focused on SSH authentication monitoring. The SIEM environment can be further enhanced to provide broader and more advanced security monitoring capabilities.

1. **Monitoring Additional Log Sources:** Logs from web servers, firewalls, Windows systems, databases, and network devices can be integrated into the SIEM environment.
2. **Advanced Detection Rules:** More detection rules can be developed to identify activities such as privilege escalation, suspicious process execution, malware-related events, and unauthorized access attempts.
3. **Automated response:** Mechanisms can be integrated to perform predefined actions when specific security alerts are generated.
4. **Threat Intelligence Integration:** External threat intelligence sources can be incorporated into enrich security events with information about known malicious IP addresses, domains, and other indicators of compromise.
5. **Machine Learning-Based Detection:** Machine learning techniques can be explored to identify unusual authentication patterns and potentially detect previously unknown threats.

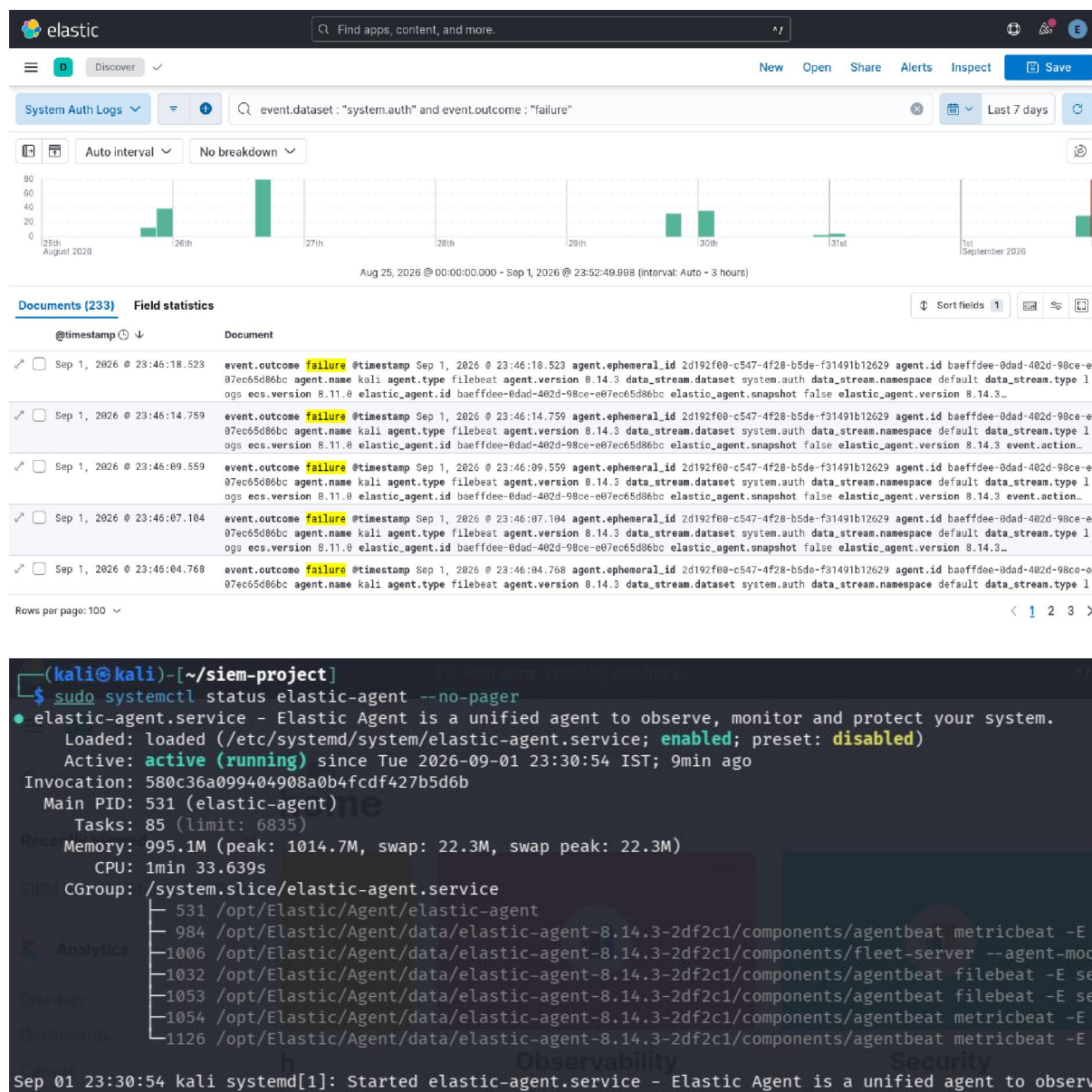
6. **Real-Time Monitoring at Larger Scale:** The architecture can be expanded from a local test environment to a distributed environment capable of monitoring multiple systems and larger volumes of security data.
7. **Improved Alert Management:** Additional alert prioritization, notification, and incident-response capabilities can be implemented to improve the overall security monitoring process.

REFERENCES

1. Elastic, *Elasticsearch Documentation*.
2. Elastic, *Kibana Documentation*.
3. Elastic, *Elastic Agent Documentation*.
4. Elastic, *Elastic Security Documentation*.
5. Docker, *Docker Documentation*.
6. Kali Linux, *Kali Linux Documentation*.
7. OpenSSH, *OpenSSH Documentation*.

APPENDIX

Appendix A – Project Screenshots



```

(kali㉿kali)-[~/siem-project]
$ sudo systemctl status ssh --no-pager
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-09-01 23:33:53 IST; 7min ago
  Invocation: f11cc90aa9f54cf7a154d7aa8ca86320
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 5412 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 5415 (sshd)
    Tasks: 1 (limit: 6835)
  Memory: 1.7M (peak: 2.8M)
     CPU: 140ms
  CGroup: /system.slice/ssh.service
          └─5415 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 01 23:33:53 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 01 23:33:53 kali sshd[5415]: Server listening on 0.0.0.0 port 22.
Sep 01 23:33:53 kali sshd[5415]: Server listening on :: port 22.
Sep 01 23:33:53 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

```

```

(kali㉿kali)-[~/siem-project]
$ ssh wronguser@localhost
wronguser@localhost's password:
Permission denied, please try again.
wronguser@localhost's password:
Permission denied, please try again.
wronguser@localhost's password:
wronguser@localhost: Permission denied (publickey,password).

(kali㉿kali)-[~/siem-project]
$ ssh wronguser@localhost
wronguser@localhost's password:
Permission denied, please try again.
wronguser@localhost's password:
Permission denied, please try again.
wronguser@localhost's password:
wronguser@localhost: Permission denied (publickey,password).

```

Select a data view

DATA VIEW .alerts-security.alerts-default,apm-transaction*,auditbeat-*,endgame-*,filebeat-*,logs-*,packetbeat-*,traces-apm*,winlogbeat-*,-*elastic-cloud-logs-*

Define your query

≡

+

🔍

event.category:"authentication" and event.outcome:"failure"

×

Set the group, threshold, and time window

WHEN count()

OVER all documents

IS ABOVE 5

FOR THE LAST 5 minutes

Set the number of documents to send

SIZE 100

☐ Exclude matches from previous runs

Add more fields to alert details

container.id ×

host.hostname ×

host.id ×

host.name ×



▶ Test query

 Copy query

Check every 

1



minute



> Advanced options

Select a data view

DATA VIEW System Auth Logs

Define your query

Set the group, threshold, and time window

WHEN count()

OVER all documents

IS ABOVE 0

FOR THE LAST 10 minutes

Set the number of documents to send

SIZE 100

☐ Exclude matches from previous runs

Add more fields to alert details

Alerts History

Columns 10 Sort fields 1 35 alerts Fields Updated 50 seconds ago   

☐	Actions	Alert Status	Last updated ↓	Started	Rule category	Rule
☐	...	Active	Sep 1, 2026 @ 23:49:19.180	Sep 1, 2026 @ 23:42:00.240	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 31, 2026 @ 22:46:54.021	Aug 31, 2026 @ 00:46:08.272	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 30, 2026 @ 22:35:11.058	Aug 30, 2026 @ 01:12:59.492	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 30, 2026 @ 01:11:56.413	Aug 30, 2026 @ 00:37:45.211	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 30, 2026 @ 00:36:39.352	Aug 30, 2026 @ 00:26:14.225	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 29, 2026 @ 23:37:29.202	Aug 29, 2026 @ 20:02:07.668	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 29, 2026 @ 20:00:39.347	Aug 29, 2026 @ 19:50:55.396	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 29, 2026 @ 19:49:53.061	Aug 29, 2026 @ 19:33:05.813	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 29, 2026 @ 19:32:02.672	Aug 29, 2026 @ 19:09:58.908	Elasticsearch query	SSH Failed Login Detection
☐	...	Recovered	Aug 26, 2026 @ 17:34:09.193	Aug 26, 2026 @ 17:03:41.343	Elasticsearch query	SSH Failed Login Detection

Rows per page: 10 < 1 2 3 4 >

Table **JSON**

Search field names		
Actions	Field	Value
	 _id	Po0vXqABfddKcIEZ
	 _index	.ds-logs-system. 0-000001
	 _score	-
	 @timestamp	Sep 1, 2026 @ 23
	 agent.ephemeral_id	2d192f00-c547-4f
	 agent.id	baeffdee-0dad-40
	 agent.name	kali
	 agent.type	filebeat
	 agent.version	8.14.3
	 data_stream.dataset	system.auth
	 data_stream.namespace	default
	 data_stream.type	logs
	 ecs.version	8.11.0
	 elastic_agent.id	baeffdee-0dad-40

 elastic_agent.snapshot	false
 elastic_agent.version	8.14.3
 event.agent_id_status	verified
 event.category	authentication
 event.dataset	system.auth
 event.ingested	Sep 1, 2026 @ 23:46:32.000
 event.kind	event
 event.module	system
 event.outcome	failure
 event.timezone	+05:30
 host.architecture	x86_64

k	host.os.codename	kali-rolling
k	host.os.family	debian
k	host.os.kernel	7.0.12+kali-amd64
k	host.os.name	Kali GNU/Linux
k	host.os.platform	kali
k	host.os.type	linux
k	host.os.version	2026.3
k	input.type	log
k	log.file.path	/var/log/auth.log
#	log.offset	53,716
k	log.syslog.appname	sshd-session
k	log.syslog.hostname	kali
k	log.syslog.procid	13696
t	message	PAM 1 more authentication failure; log name= uid=0 euid=0 tty=ssh ruser= rhost= t=::1
k	process.name	sshd-session
k	related.hosts	kali
k	tags	system-auth